



Background colouration influences body colour segregation in mollies

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(Received 1 July 2000, Accepted 8 June 2001)

When given a choice of swimming against a black background near an empty black end compartment or swimming against a white background near an empty white compartment, black morph mollies *Poecilia latipinna* spent significantly more time against the black background. The presence of a shoal (either white or black mollies) in the black compartment increased the preference of the test fish for the black background. However, black molly test fish spent significantly less time against a black background if a black shoal was present in the white compartment than in assays in which a white shoal was in the white compartment. In assays where test fish were given the choice between two black shoals, test fish spent significantly more time near the black shoal on the black background. Similarly, when given the choice between two white shoals, test fish spent significantly more time against the black background. The results suggest that the presence of cryptic backgrounds alter shoalmate choices in fish that assort according to body colouration.

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Key words: shoaling behaviour; phenotypic assortment; body colour; background colouration.

INTRODUCTION

Individuals in many species of teleosts gain fitness advantages in terms of protection from predators by joining groups known as shoals (Godin, 1997). These advantages can be as simple as numerical dilution in which the probability that any one individual within a shoal will be captured decreases as shoal size increases (Morgan & Godin, 1985), or as complex as information transfer between individuals to increase the corporate vigilance of the group and allow earlier detection of predators (Magurran *et al.*, 1985; Godin *et al.*, 1988; Magurran, 1990; Pitcher & Parrish, 1993). Furthermore, fishes in shoals may benefit from true schooling behaviour, involving complex evasive actions and coordinated manoeuvres by individuals within the group (Breder, 1959; Pitcher & Parrish, 1993).

Another important antipredator benefit associated with shoaling is the confusion effect in which a predator, faced with a group of moving targets, has difficulty singling out and capturing an individual fish (Neill & Cullen, 1974; Milinski, 1977; Ohguchi, 1981). The confusion effect appears to be related to the high degree of phenotypic similarity typical of many shoals. The visual image of many similar fish may make it difficult for a predator to target a specific

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individual. Conversely, in a related phenomenon known as the oddity effect, the presence of phenotypically odd individuals within the shoal reduces the sensory confusion of a potential predator, increasing the chances of a successful kill (Landeau & Terborgh, 1986; Theodorakis, 1989).

Selective pressures imposed by the oddity effect may explain why fishes choose to shoal with individuals of similar size (Ranta *et al.*, 1992; Krause, 1994; Krause & Godin, 1994; Peuhkuri, 1997), parasite load as evidenced by conspicuous spots (Krause & Godin, 1996), and body colouration (McRobert & Bradner, 1998). However, phenotypic similarity may not be the only factor involved in shoalmate choice. Recent studies suggest that patterns of partner choice may result from a number of associated selective pressures. For example, size assortment may be driven by both the oddity effect and by pressures resulting from food competition (Peuhkuri, 1997). Small fish choosing to shoal with larger fish may not be able to compete for food. Furthermore, some shoals may assort phenotypically in certain situations but not in others. For example, mixed species shoals may segregate into groups of conspecifics when threatened by predators, enhancing benefits associated with coordinated manoeuvring and predatory confusion (Wolf, 1985; Allan & Pitcher, 1986). Sub-groups of stoplight parrotfish *Sparisoma viride* (Bonnaterre) are often found foraging within larger groups of heterospecifics, with whom they appear highly conspicuous. However, the conspicuous individuals abandon the larger group and seek hiding places during predatory encounters (Wolf, 1985). Therefore, different association choices may be made under different environmental situations.

In a previous study (McRobert & Bradner, 1998), black and white colour morphs of mollies *Poecilia latipinna* (Lesueur) were tested for association choices based on visual cues. The results showed that these mollies chose to shoal with individuals of similar colouration, presumably increasing fitness through the confusion effect. This paradigm provided a simple (literally black and white) model for the study of association choices under laboratory conditions. In the present study black and white mollies are used to test the interactions between potential shoalmate colouration and background colouration to test the influence of environmental conditions on association choices.

MATERIALS AND METHODS

TEST SUBJECTS

Black and white mollies, obtained from World Wide Aquarium in Philadelphia, were housed in groups of 40 in four separate glass holding tanks (91 × 46 × 33 cm). Black and white mollies were housed separately (as they were in the pet store), and opaque partitions prevented black and white mollies from seeing each other. Aquaria were maintained under fluorescent lighting, on a 12L : 12D cycle, at 23° C. Marine-grade salt was added to each aquarium (5 g l⁻¹) to reduce the chance of fungal infection. All fish were fed a commercial flake food once per day.

TEST TANKS

Test tanks (Fig. 1) were constructed by dividing aquaria (74 × 31 × 31 cm) into three separate compartments (McRobert & Bradner, 1998). The two end compartments (18 × 31 × 31 cm) were separated from the central compartment by clear glass partitions.

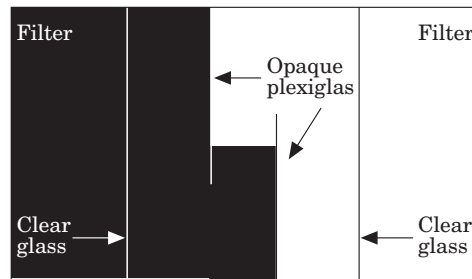


FIG. 1. Diagram of test tank used to study the interaction between preferences for background colour and shoalmate colour in mollies.

Black plastic sheeting was placed over the outside of each test tank such that one-half of the back wall, one side and one-half of the bottom was covered. White plastic sheeting covered the other half of each test tank. In this way, target fish in one end compartment swam against a white background, while target fish in the other end compartment swam against a black background. The front side of each tank was left uncovered to allow observers to view the fish. The central compartment of each test tank was further subdivided into left and right sides using opaque Plexiglas partitions (25×18 cm). These partitions allowed test fish to move from one side of the central compartment to the other but prevented a fish on one side from seeing either the target shoal or the background colour in the end compartment on the other side. There was no exchange of water between the three compartments. All test tanks included heaters and filters.

BEHAVIOURAL ASSAYS

Tests were conducted twice each day (late morning and early afternoon) from Monday to Friday, during January to August, 1999. Each test session consisted of a 600 s observation period during which the time the test fish spent on each side of the central compartment was recorded. A week of tests (10 test sessions on an individual test fish, or 6000 s) was defined as a batch. Fish were removed from test tanks following the afternoon tests on Friday, and were replaced with fish for the next week's experiments. Fish were chosen at random, with the exception that an attempt was made to select equal numbers of males and females as test fish for each experiment. The fish were given the weekend to acclimate to the test tanks prior to the first test session on Monday. Each test fish was used only once.

For each behavioural assay, a single test fish was placed into the central compartment. Only black mollies were used as test fish due to their pronounced preference for the black backgrounds. White mollies failed to demonstrate a preference for either background colour utilized in this experiment. The experiment followed a 3×3 factorial design with shoal composition in the black end compartment (no shoal, black shoal, white shoal), and shoal composition in the white end compartment (no shoal, black shoal, white shoal) as main factors. A shoal was defined as a group of four target fish. Twelve batches for each assay were conducted. An equal number of sessions were run with the left and the right half of the test tanks covered in black plastic sheeting to eliminate the risk of 'tank effects'.

STATISTICAL ANALYSIS

For each batch (10 test sessions on an individual fish), the total time spent by the test fish on the black and white sides of the central compartment was calculated for statistical analysis. In this way, a single data point was generated for each test fish. One-sample *t*-tests, compared with a null expectation (0.5 total time) were used to compare time spent on the black side of the central compartment to time spent on the white side of the central compartment for each individual assay. Batch totals were grouped according to

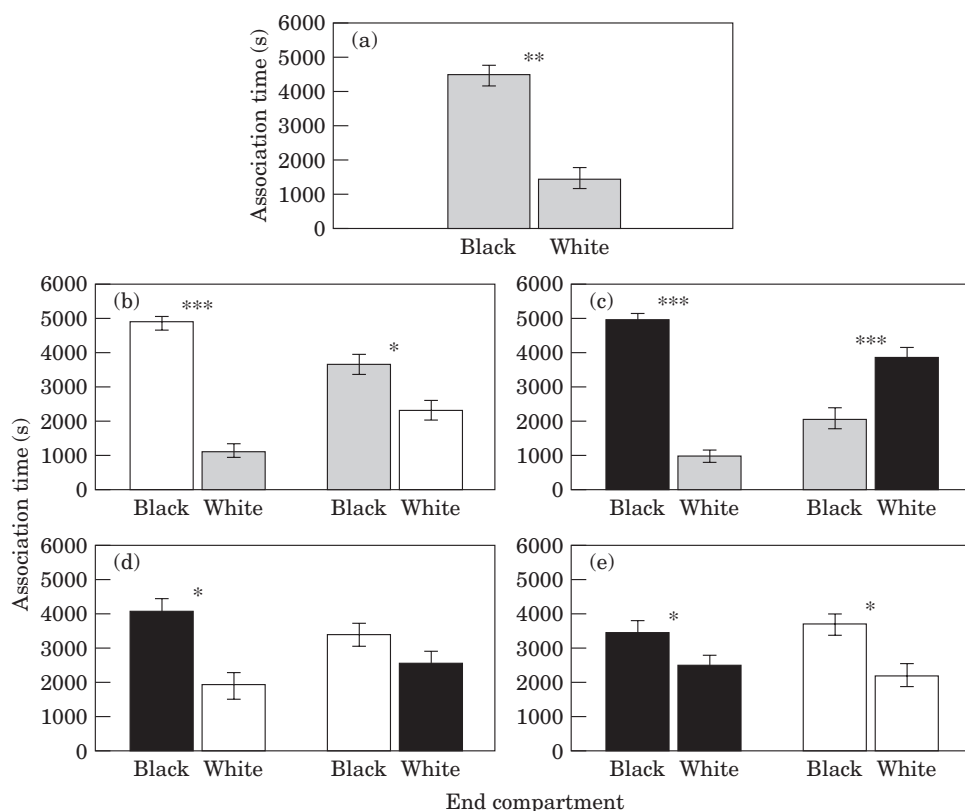


FIG. 2. Mean ($\pm$ S.E.) time spent by black test fish on the black and white coloured sides of the central compartment using (a) both end compartments empty; (b) one white target shoal; (c) one black target shoal; (d) two shoals of different colour; (e) two shoals of the same colour. Test fish were given choices between black and white end compartments that were either empty ($\square$) or contained four white mollies ($\square$) or four black mollies ($\blacksquare$). Significant differences between time spent on the black and white sides of the central compartment were determined with one-sample *t*-tests for all possible shoal combinations ($n=12$). *** $P<0.001$, ** $P<0.01$, * $P<0.05$.

treatment condition and analysed using a 2-factor ANOVA on main factors and their interactions. The ANOVA was followed by Tukey HSD multiple comparisons where appropriate ($q_{0.05,99,3}=3.36$).

RESULTS

Black molly test fish showed a significant preference for the black side of the central chamber when both end compartments were empty [assay 1, Figs 2(a) and 3]. When given a choice between a white target shoal and no shoal (assays 2 and 3), test fish spent significantly more time on the black side of the central chamber, even when the white target shoal was in the white end compartment [Figs 2(b) and 3]. However, test fish spent significantly more time on the side of the central compartment closest to a black shoal when the other end compartment was empty (assays 4 and 5), regardless of background colour [Figs 2(c) and 3]. The test fish, given the choice between a black shoal and a

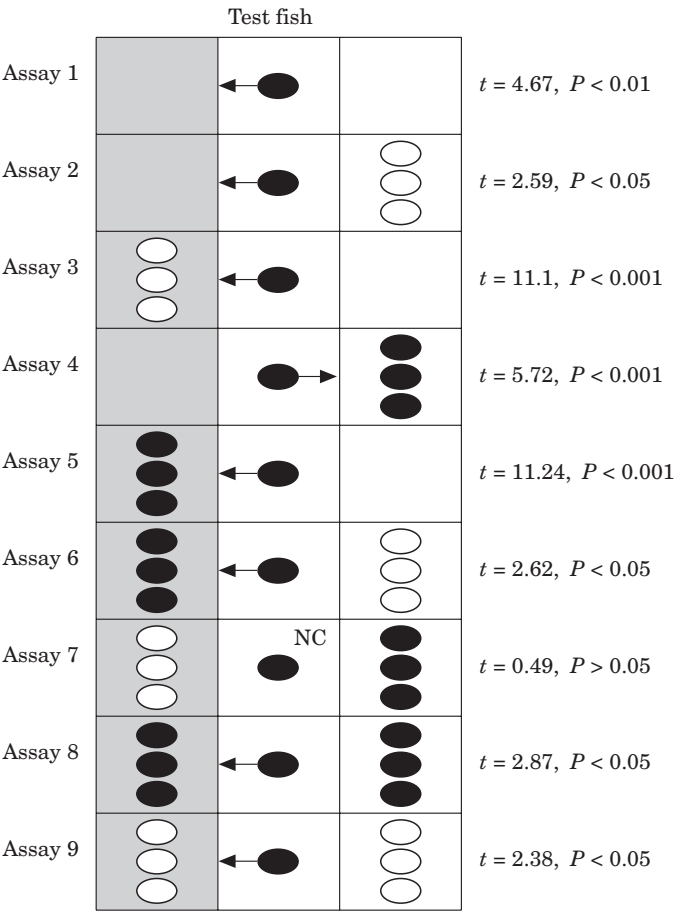


FIG. 3. Diagram representing the results of the study. Black coloured side (left column) of test tank is shaded. White coloured side (right column) of test tank is unshaded. ●, Black molly; ○, while molly. Arrows represent a significant choice by test fish. NC, No significant choice by test fish.

white shoal (assays 6 and 7), chose to spend significantly more time on the side of the central compartment closest to the black shoal only when the black shoal was in the black end compartment [Figs 2(d) and 3]. There was no significant difference between the time spent by the test fish on either side of the central compartment when the target shoals were in end compartments of opposite colour. When given the choice between two black shoals (assay 8), test fish spent significantly more time on the black side of the central chamber [Figs 2(e) and 3]. Likewise, in assay 9, the test fish spent significantly more time on the black side of the central chamber when given the choice between two white shoals.

Both the shoal composition in the black end compartment and the shoal composition in the white end compartment had significant effects on the amount of time the test fish spent on the black side of the central chamber (Table I). *Post hoc* multiple comparisons revealed that the test fish spent more time on the black

TABLE I. Results of 2 factor ANOVA describing the influence of shoal composition in the black (SCB) and white (SCW) end compartments, and their interaction, on time spent by black test fish on the black coloured side of the central compartment

Source	Sum of squares	d.f.	Mean squares	F-ratio	P
SCB	11 779 810	2	5 889 904·861	5·690	<0·001
SCW	57 184 321	2	28 592 160·528	27·622	<0·01
SCB × SCW	5 704 207	4	1 426 051·681	1·378	0·247
Total	177 146 720	107	1 655 576·822		

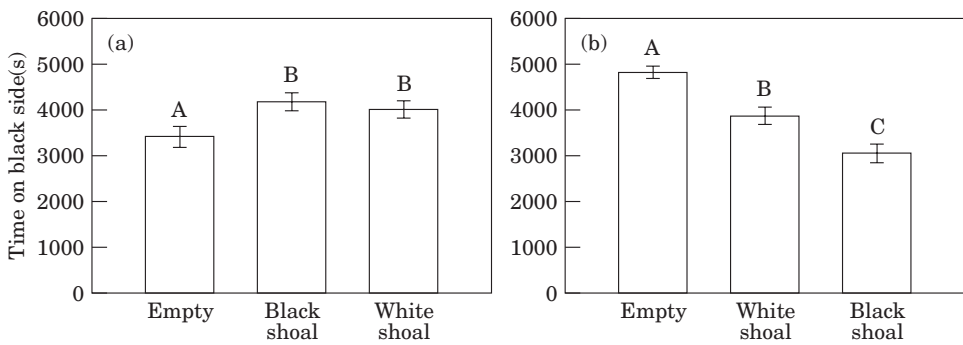


FIG. 4. Mean ($\pm$ s.e.) time spent by black test fish on the black side of the central compartment grouped by shoal composition in the black end compartment (a) and shoal composition in the white end compartment (b). $n=36$ for all data points. Bars with similar letters are not significantly different from each other.

side of the central chamber when a black shoal ($q=4\cdot56$, $n=36$, $P<0\cdot05$) or a white shoal ($q=3\cdot49$, $n=36$, $P<0\cdot05$) was present in the black end compartment than when the black end compartment was empty [Fig. 4(a)]. There was no difference in the time spent by the test fish on the black side of the central chamber when either a black or a white target shoal was present in the black end compartment [$q=1\cdot06$, $n=36$, N.S., Fig. 4(a)].

Black test fish spent more time on the black side of the central chamber when the white end compartment was empty than when the white end compartment contained a black shoal ($q=10\cdot50$, $n=36$, $P<0\cdot05$) or a white shoal [$q=5\cdot60$, $n=36$, $P<0\cdot05$, Fig. 4(b)]. Additionally, the black test fish spent more time on the black side of the central compartment when the white end compartment contained a black target shoal than when it contained a white target shoal ($q=4\cdot89$, $n=36$, $P<0\cdot05$, Fig. 4(b)).

Regression analysis of the time spent in the black side of the central compartment on individual test periods failed to indicate any discernible trends of behaviour modification for any individual test fish. Additionally, comparisons between the behaviour of test fish during the first half of the week and during the last half of the week supported these conclusions.

DISCUSSION

The importance of background colouration is shown by the fact that the black test fish chose the black background over the white background in almost every assay. While the aim of this study was not to determine why black test fish chose black backgrounds over white ones, one explanation may be that the black background offered cryptic advantages over the white background (against which black mollies are highly conspicuous). Another possibility was that the black background was simply darker and attracted the black mollies as a potential hiding place. In either case, these findings are supported by field observations showing that fish schools prefer to swim against matching backgrounds (Breder, 1959).

With regard to the interaction between background colouration and shoal colouration, black test fish showed a significant preference for the black background in all assays in which a black shoal was located in the black end compartment. Under these circumstances the potential benefits of shoaling, including feeding and mating benefits, would be augmented by the confusion effect. When presented with a shoal of white mollies, black test fish again chose the black background, regardless of the location of the white shoal. This indicates that the benefits of swimming against a background of similar colouration may override the benefits, and potential dangers, of shoaling with dissimilarly-coloured individuals. When the white shoal was swimming against the black background, the test fish could gain the benefits of swimming against the black background as well as benefit from joining a shoal. In this situation the potential dangers of shoaling with dissimilar individuals may be reduced by the possibility of abandoning the shoal in the presence of a predator, then blending into the background for protection. This suggestion is supported by a field study in which odd individuals in a mixed-species shoal abandoned the group and took shelter in hiding places during predatory encounters (Wolf, 1985).

The only cases in which the test fish did not show a significant preference for the black side of the test tank were assays 4 and 7. In assay 4, black test fish chose a black shoal over swimming alone against a black background. Here the benefits of shoaling appear to override the chance to be concealed against the background. However, in assay 7, test fish showed no significant preference for either a black shoal swimming against a white background or a white shoal swimming against a black background. Within this situation it appears that both choices may provide benefits.

The results of this study indicate that the decisions involved in shoalmate choice are complex, involving an interplay between phenotypic components of the shoal as well as characteristics of the surrounding environment. It is important to note that the findings reported here are limited to some extent by the use of captive-bred animals. Black and white mollies were chosen for this study because of their distinct colouration and their pronounced tendency to assort according to body colour (McRobert & Bradner, 1998). Unfortunately, these animals have not been exposed to natural selective pressures, which raises questions about their ability to represent animals in the wild. However, the ability to discriminate between potential shoalmates on the basis of phenotypic features plays such an important role in the lives of fishes that it is

likely to be highly conserved and could still be present in captive-bred fishes. This suggestion is supported by previous studies on convict cichlids *Cichlasoma nigrofasciatum* (Günther) and zebra danios *Brachydanio rerio* (Hamilton-Buchanan), which suggest that the tendency to sort according to body colour is both common and highly conserved in captive bred species (S. P. McRobert, unpubl. data). It will be important in future studies to replicate these findings to determine whether other fish species choose shoalmates on the basis of body colouration and determine whether these choices are influenced by the colour of the background. This study has established the basis for such future studies by the describing the interplay between shoalmate colouration and background colouration in shoalmate choice.

We thank J. Mindell for assistance with the statistical analyses and P. Tefft and G. Ettl for comments on an earlier draft of this manuscript. We also thank two anonymous referees for their constructive comments.

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